6.11 Approximating Integrals: Trapezoidal Sums, the Midpoint Rule, Simpson's Rule

Name _____ Period ____ Date _____

AP Calculus 6.11 · Student sheet

I can…

- approximate an integral with a trapezoidal sum, including from a table.
- use the Midpoint Rule and Simpson's Rule.
- tell when an approximation is an overestimate or an underestimate.

Vocabulary · fill in the missing word

Trapezoidal sum — _____

Midpoint Rule — a Riemann sum using the midpoint of each _____

Simpson's Rule — $(\Delta\frac{x}{3})[f(x_0) + 4f(x_1) + 2f(x_2) + \ldots + 4f(x_{n-1}) + f(x_n)]$, n _____

Part 1 · Rectangles vs. trapezoids

$\int_0^2 x^2\,dx = 8/3 \approx 2.67$, n = 2.

Method	Approximation
Left	
Right	
Trapezoids	

Part 2 · Notes and examples (fill in each note, then work the example)

Trapezoidal sum: T is the _____ of the left and right Riemann sums.

Example 1

Approximate $\int_0^2 x^2\,dx$ with a trapezoidal sum, n = 4.

Unequal widths: add (width)(average of the two _____) for each trapezoid.

Example 2

x: 0, 2, 5, 7, 10 and f(x): 3, 5, 6, 4, 8. Approximate $\int_0^{10} f(x)\,dx$ with trapezoids.

Simpson's Rule: coefficients 1, 4, 2, 4, …, 4, 1, and n must be _____.

Example 3

Approximate $\int_0^2 x^2\,dx$ with Simpson's Rule, n = 4.

Concavity: if f is concave up, the trapezoidal sum is an _____.

Example 4

f is concave up on [a, b]. Is a trapezoidal sum an overestimate or an underestimate?

Part 3 · Partner practice

Partner 1

Approximate $\int_0^4 x^2\,dx$ with the Midpoint Rule, n = 2.

Partner 2

(AP® Practice) The area under $y = x^3$ from 0 to 4 is approximated with a trapezoidal sum using four subintervals of equal width.

The approximation is

(A) 34 (B) 68 (C) 50 (D) 136